

Morphological Image Processing

Unit- 5

Morphology and Mathematical Morphology

- The word *morphology* Commonly denotes a branch of biology that deals with the form and structure of animals and plants.
- We use here the same word here in the context of *mathematical morphology* as a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons, and the convex hull.
- Mathematical morphology is a branch of image processing that has been successfully used to provide tools for representing, describing, and analyzing shapes in figures.
- In addition to providing useful tools for extracting image components morphological algorithms have been used for pre- or post processing the images containing shapes of interest such as filtering, thinning, and pruning.

Principal of Mathematical Morphology

- To extract the geometrical and topological information from an unknown set of image through transformations using another, well defined, set known as structuring element (SE).
- Note:
- In morphological image processing, the design of SEs, their shape and size, is crucial to the success of the morphological operations that use them.

Fundamental Concept and Operations

- The language of mathematical morphology is set theory and its standard operations: *union*, *intersection* and *complement*. Sets in mathematical morphology represents objects in an image.
- For example: the set of all white pixels in a binary image is a complete morphological description of the image.
- In addition to the basic set definitions, the concept of set reflection and translation are used extensively in morphology.
- Morphological Theory views on binary image as a set of foreground (one valued) pixel and elements of which are in Z^2 (Cartesian product) i.e. pair of elements.
- Set operations such as union, intersection and complement can be applied directly to binary image sets.

Union of Images

- If A and B are binary images then

$C = A \cup B$ is binary image where, a pixel in C is a foreground pixel if either both the corresponding pixel in A and B are foreground pixel. or

- The function C is given by two view. They are:

i) First view

$$C(x, y) = \begin{cases} 1, & \text{if either } A(x, y) \text{ or } B(x, y) \text{ is } 1 \\ & \text{or} \\ & \text{both are } 1 \\ 0, & \text{otherwise} \end{cases}$$

ii) Second view

$$C = \{ (x, y) : (x, y) \in A \text{ or } (x, y) \in B \text{ or } (x, y) \in (A \text{ and } B) \}$$

where A and B are Binary Images where the elements of A and B are 1-valued.

Intersection of Two images

- $C(x, y) = \begin{cases} 1 & \text{if } A(x, y) \text{ and } B(x, y) \text{ are both } 1 \\ 0 & \text{otherwise} \end{cases}$

Complement of Image

$$A^c = \{z \mid z \notin A\}$$

Difference of Two Images

$$A - B = \{z \mid z \in A, z \notin B\} = A \cap B^c$$

Translation

Let A be a set (of pixels in a binary image) and $w = (x, y)$ be a particular coordinate point. The translation of set A by point w is denoted by A_w and defined as:

$$A_w = \{c \mid c = a + w, \text{ for } a \in A\}$$

Reflection

The reflection of set A relative to the origin of a coordinate system, denoted by $\hat{A}$, is defined as

$$\hat{A} = \{z \mid z = -a, \text{ for } a \in A\}$$

The set operation can be performed on binary images using following logical operations:

S. No	Set Operation	Logical Operation for Binary Images	Name
1	$A \cup B$	$A \mid B$	OR
2	$A \cap B$	$A \& B$	AND
3	A^c	$\sim A$	NOT
4	$A-B$	$A \& \sim B$	DIFFERENCE

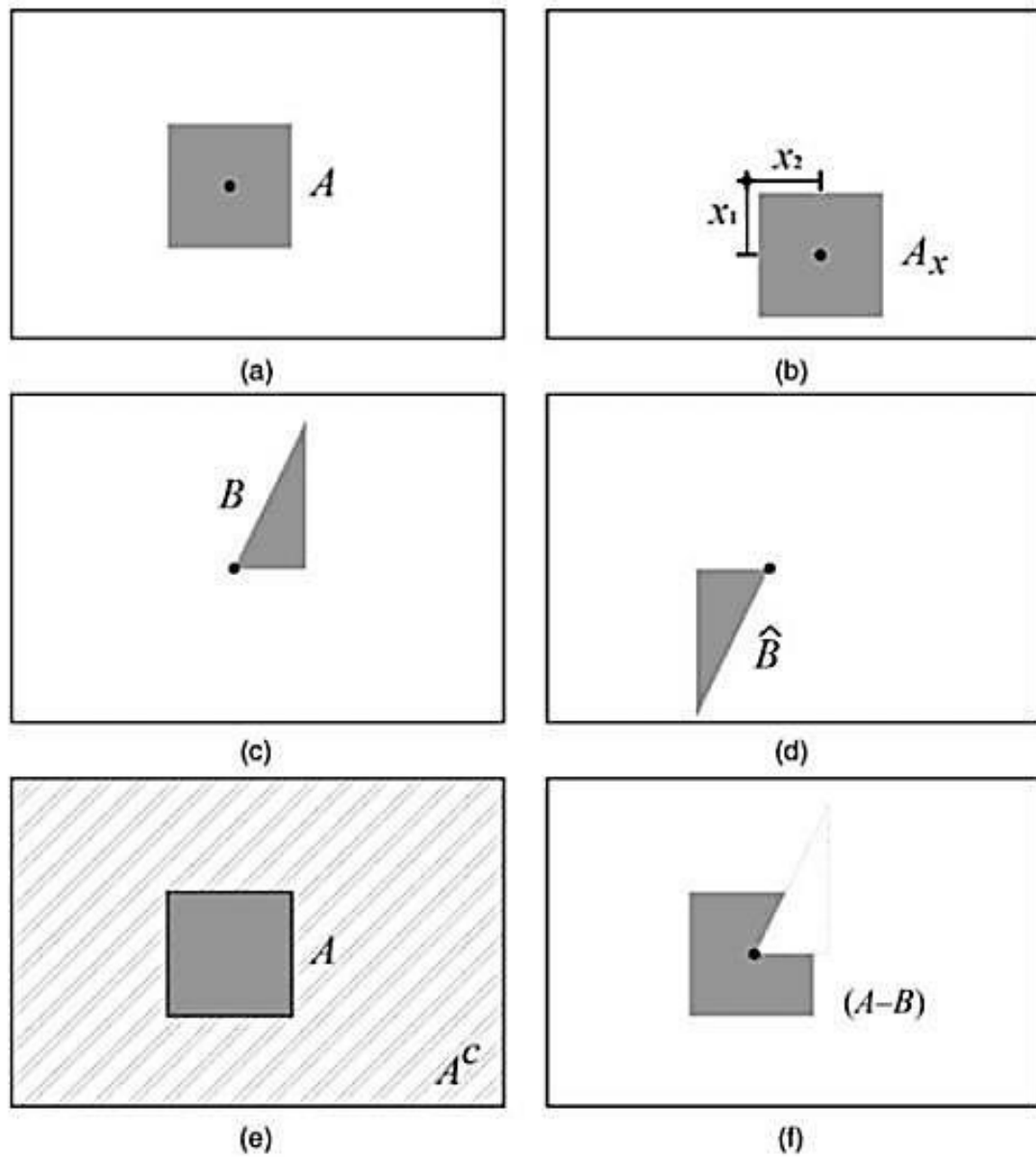


FIGURE Basic set operations: (a) set A ; (b) translation of A by $x = (x_1, x_2)$; (c) set B ; (d) reflection of B ; (e) set A and its complement A^c ; (f) set difference $(A - B)$.

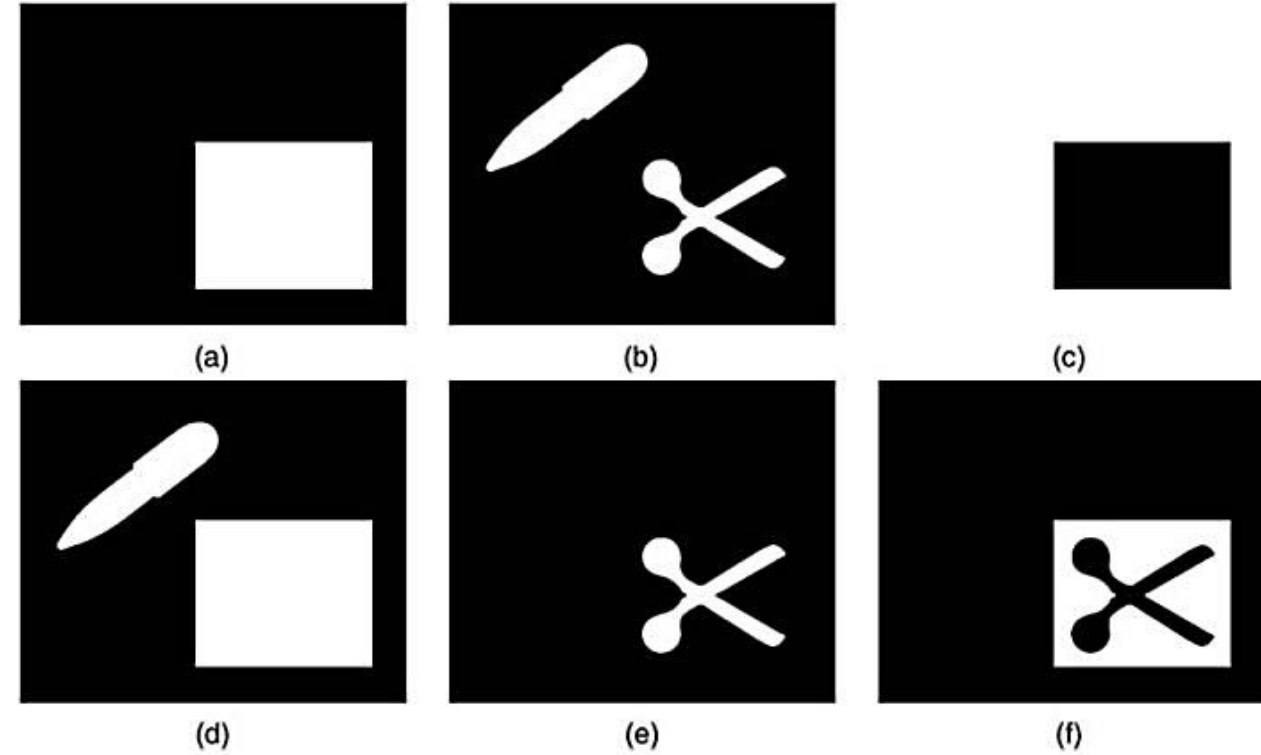


FIGURE Logical equivalents of set theory operations: (a) Binary image (A); (b) Binary image (B); (c) Complement (A^c); (d) Union ($A \cup B$); (e) Intersection ($A \cap B$); (f) Set difference ($A - B$).

Structuring Elements (SEs)

- It is small sets of sub-images used to probe an image under study for properties of interest.
- A structuring element is a matrix that identifies the pixel in the image being processed and defines the neighborhood used in the processing of each pixel.
- It is usually represented as a small matrix (i.e. rectangular array) whose shape and size impact the results of applying a certain morphological operator to an image.
- In SE, the origin is specified by a black dot. In general, place at its center of gravity.
- Although, a SE can have any shape, its implementation requires that it should be converted to a rectangular array. For each array, the shaded squares correspond to the members of the SE whereas the empty squares are used for padding only.

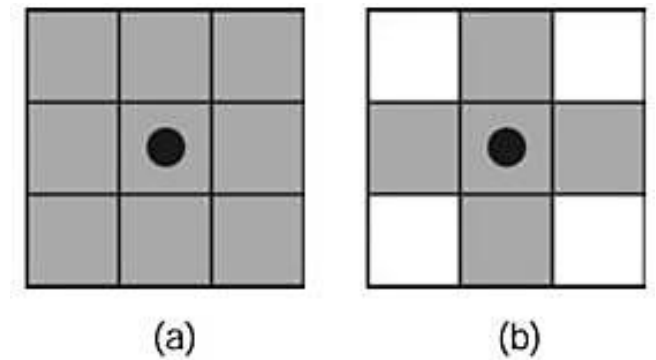


Figure: Examples of SE:
(a) Square; (b) cross

Fundamental Morphological Operations

- There are two fundamental morphological operations upon which all other operations and algorithms built. They are:
 1. Dilation, and
 2. Erosion

Dilation

- Dilation is a morphological operation whose effect is to “grow” or “thicken” objects in a binary image.
- the extent and direction of this thickening are controlled by the size and shape of the structuring element.
- Graphically, structuring elements can be represented either by a matrix of 0’s and 1’s or as a set of foreground (1-valued) pixels.
- So, the origin of the structuring element must be clearly identified by using the both representation.

Dilation

- Mathematically, the dilation of a set A and B , denoted by $A \hat{\boxplus} B$, is defined as $A \hat{\boxplus} B = \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$
- This equation is based on reflecting B about its origin, and shifting this reflection by z .
- The dilation of A by B then is the set of all displacements, such that B and A overlap by at least one element. Based on this interpretation, the above equation can be written equivalently as

$$A \hat{\boxplus} B = \{z \mid [(\hat{B})_z \cap A] \subseteq A\}$$

where we assume B is a structuring element and A is the set (image objects) to be dilated.

Dilation

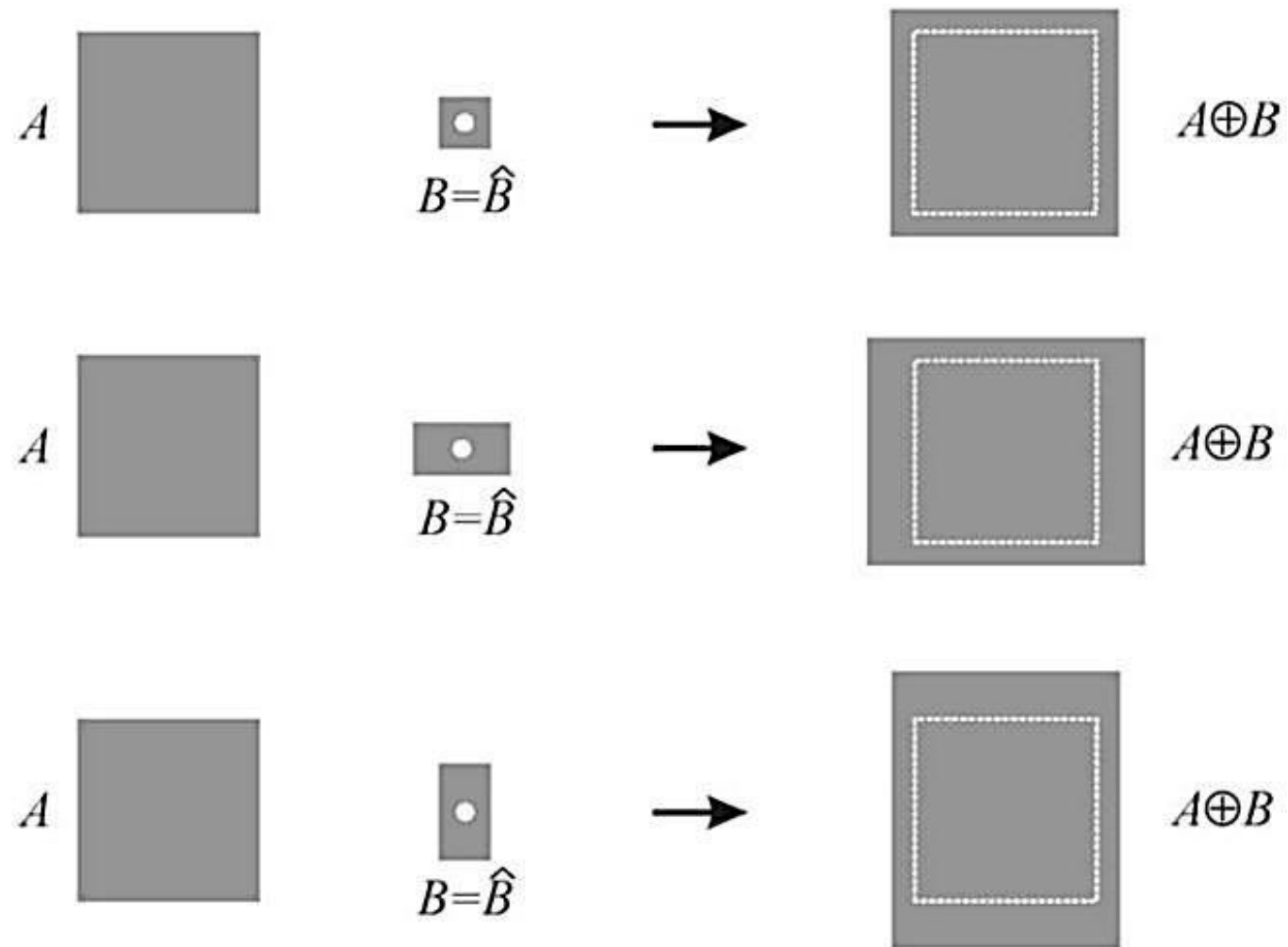


FIGURE Example of dilation using three different rectangular structuring elements.

Properties

- Dilation is Associative

$$A \hat{\cup} (B \hat{\cup} C) = (A \hat{\cup} B) \hat{\cup} C$$

- Dilation is Commutative

$$(A \hat{\cup} B) = (B \hat{\cup} A)$$

Erosion

- Erosion is a morphological operation whose effect is to “shrink” or “thin” objects in a binary image. The direction and extent of this thinning is controlled by the shape and size of the structuring element.
- Mathematically, the erosion of a set A by B , denoted by $A \ominus B$, is defined as

$$A \ominus B = \{z \mid (B)_z \cup A^c \neq \emptyset\}$$

- In words, this equation indicates that the erosion of A by B is the set of all points z such that B , translated by z , is contained in A .

Erosion

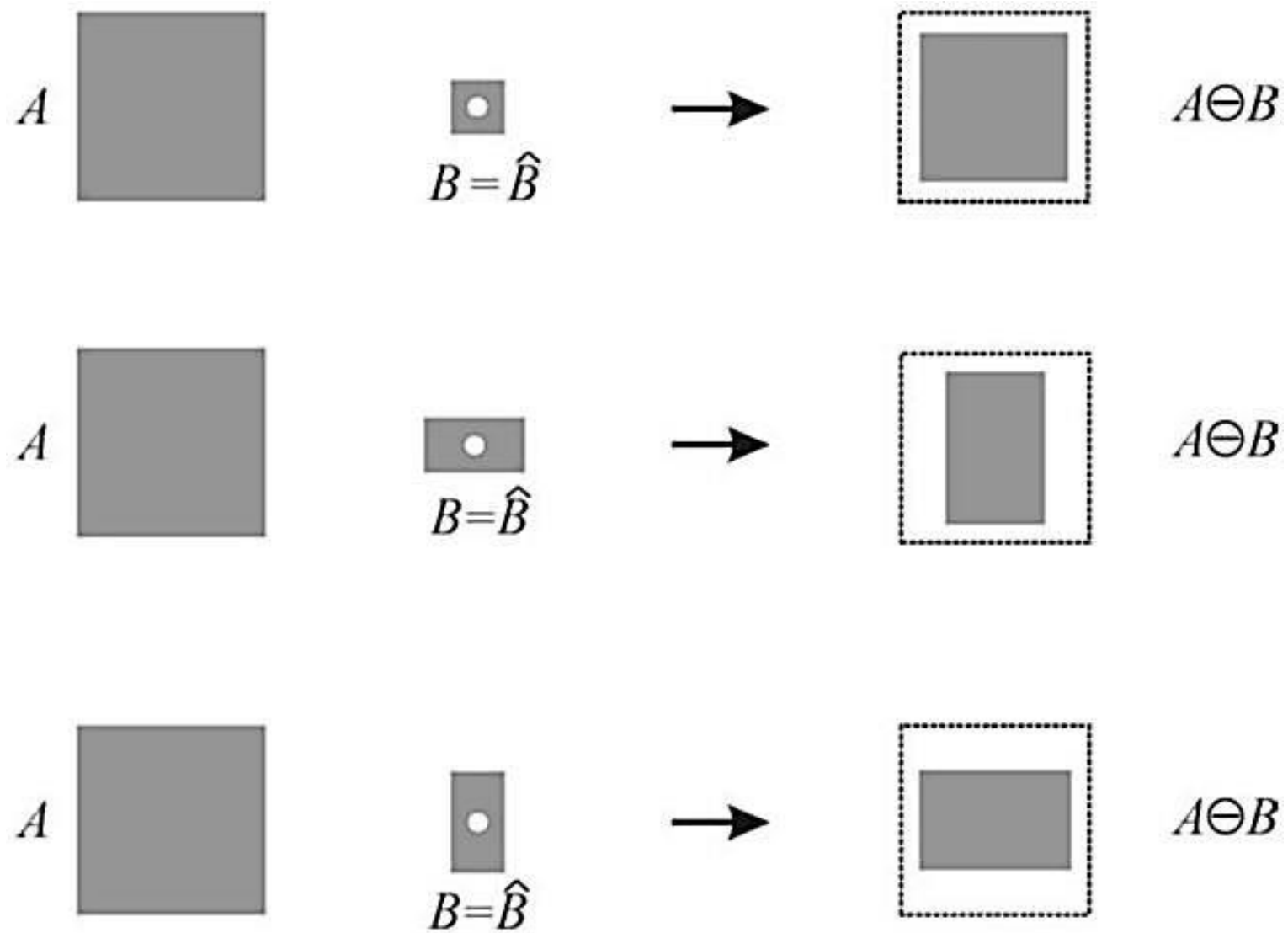


FIGURE Example of erosion using three different rectangular structuring elements.

Duality

- Erosion and Dilation are duals of each other with respect to set complementation and reflection. That is,

$$(A \ominus B)^c = A^c \hat{\boxplus} B$$

$$(A \hat{\boxplus} B)^c = A^c \ominus B$$

Compound Operations

- Operations that combine the two fundamental operations (erosion and dilation).
 1. Opening
 2. Closing

Opening

- The morphological opening of set A by B is denoted by $A \circ B$, and defined as the erosion of A by B followed by the dilation of the result by B. Mathematically,

$$A \circ B = (A \ominus B) \oplus B$$

Alternatively, the opening operation can be expressed using set notation as

$$A \circ B = \dot{\bigcup} \{ (B)_z \mid (B)_z \subseteq A \}$$

where $\dot{\bigcup} \{ . \}$ represents the union of all sets within the curly braces, and the symbol $\subseteq$ means “is a subset of.”

- $A \circ B$ is the union of all translation of B that fit entirely within A

Geometrical Interpretation of opening operation

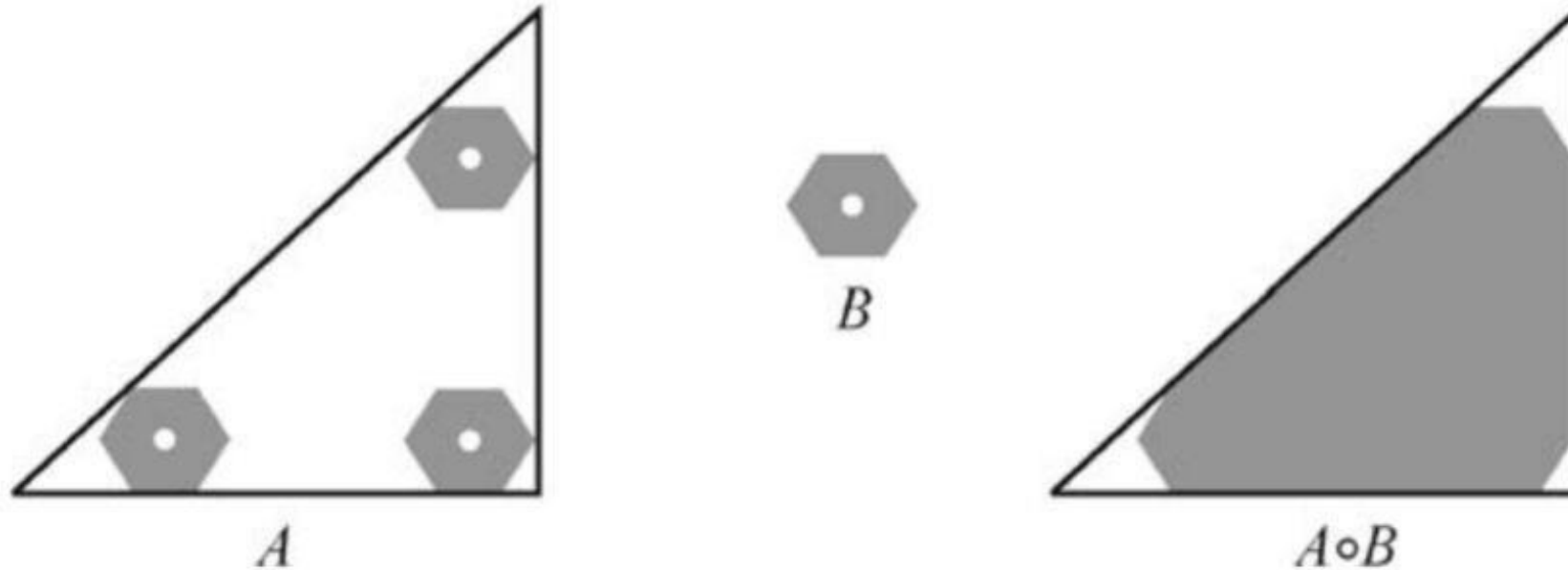


FIGURE 1 Geometric interpretation of the morphological opening operation.

Example of opening operation

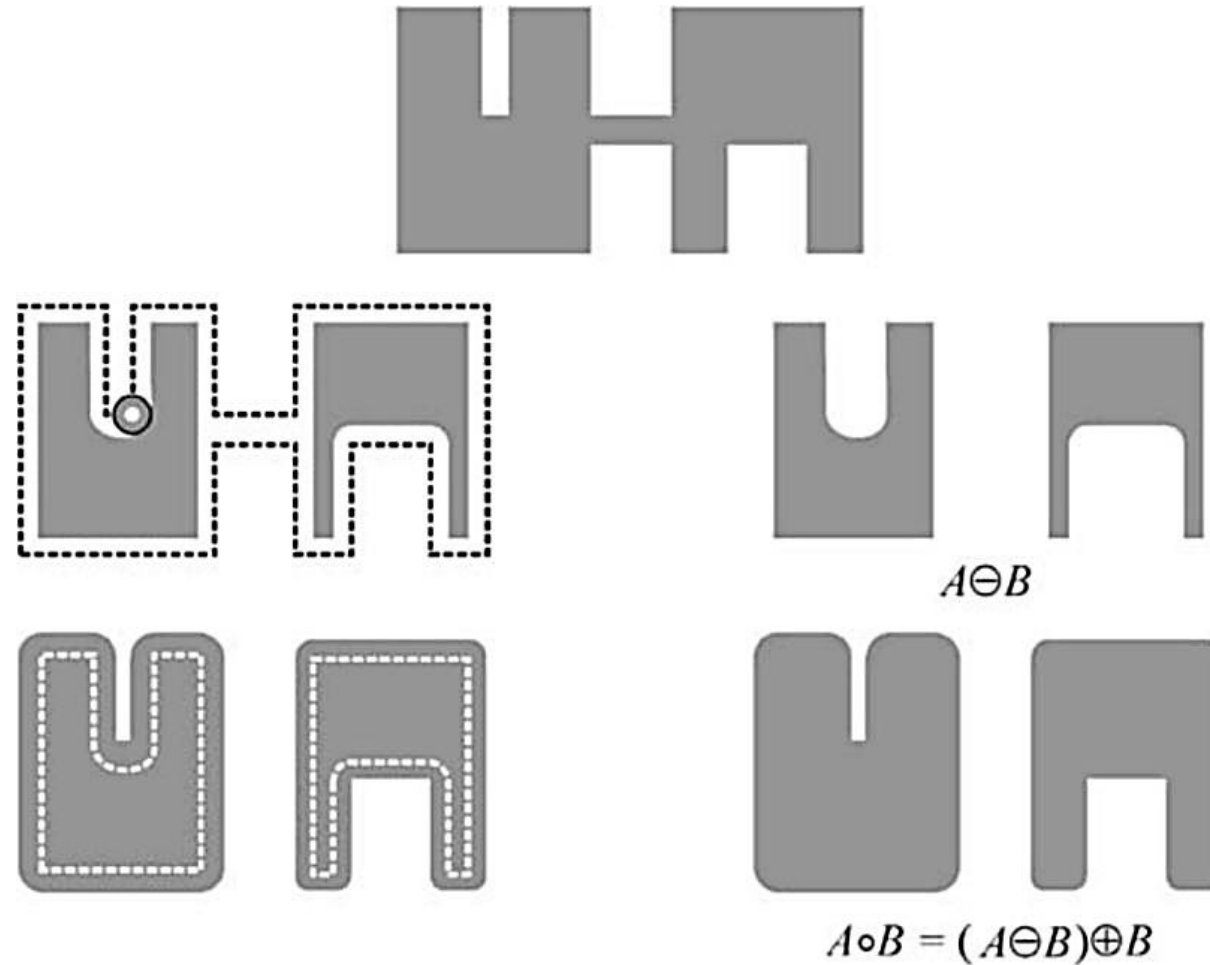


FIGURE Example of morphological opening.

Uses of opening operation

- Morphological opening is typically used to remove thin protrusions from objects and to open a gap between objects connected by a thin bridge without shrinking the objects (as erosion could have done)
- It also causes a smoothening of the object's contour.

Closing

- The morphological closing of A by B is denoted by $A \bullet B$ and defined as the dilation of A by B and followed by an erosion of the result by B . Mathematically,

$$A \bullet B = (A \oplus B) \ominus B.$$

- Geometrical interpretation: $A \bullet B$ is the complement of the union of all translations of B that do not overlap A .

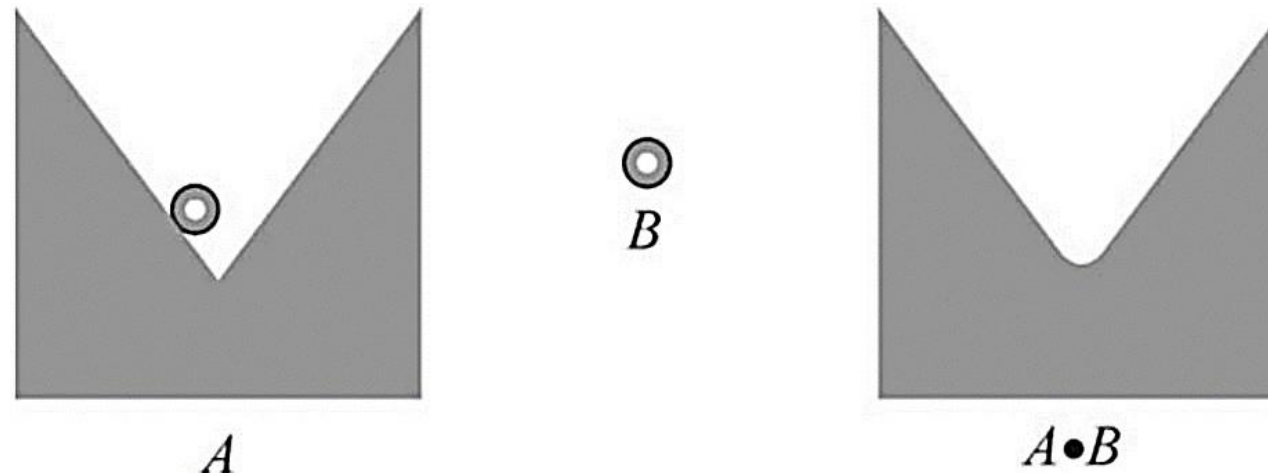


Figure: Geometrical Interpretation of closing operation

Uses of closing operation

- Morphological closing is typically used to fill small holes, fuse narrow breaks, and close thin gaps in the objects within an image, without changing the objects' size (as dilation have would have done)
- It also causes smoothening of the object's contour.

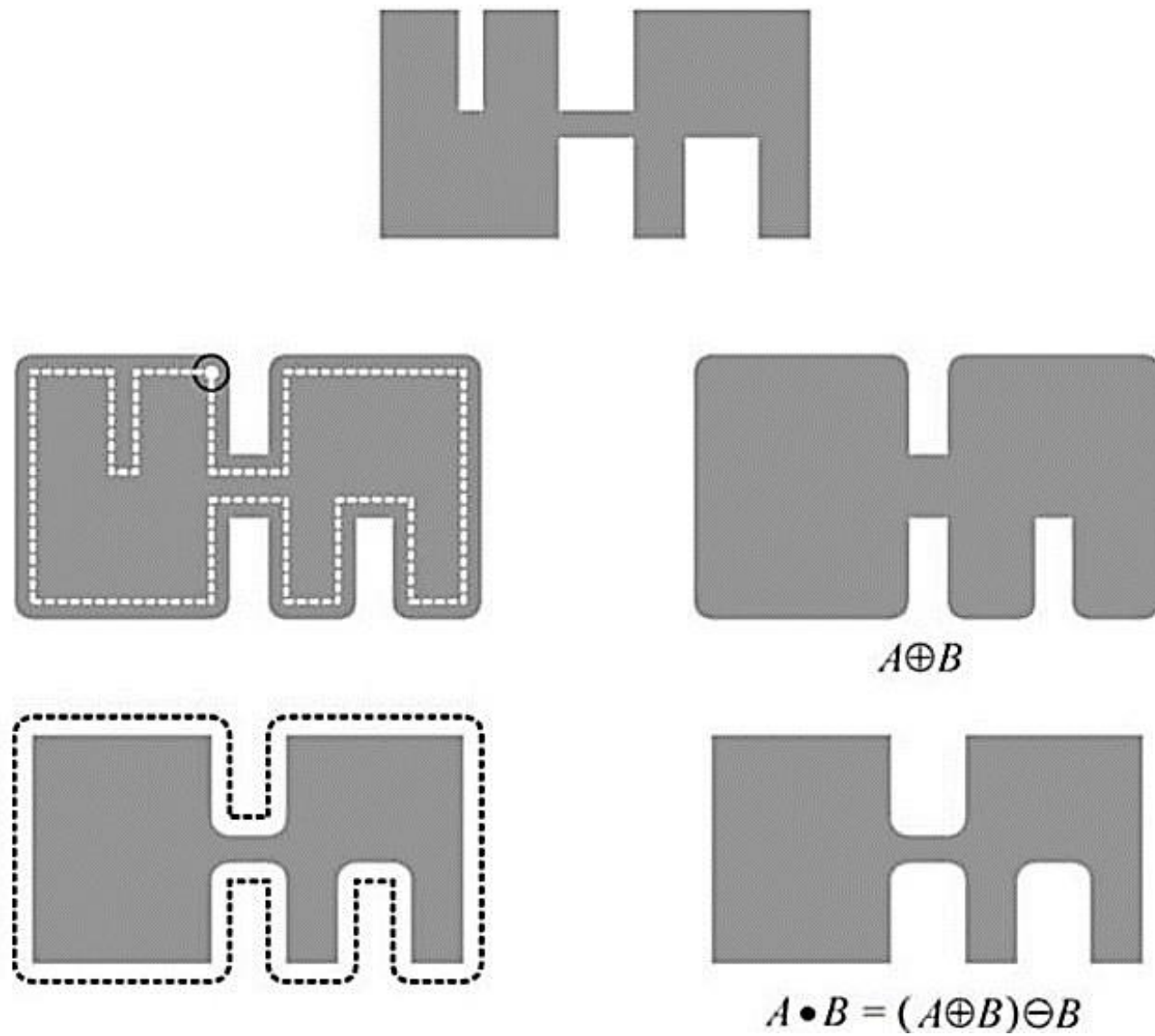


Figure: Examples of morphological closing operation

Duality

- Closing is the dual operation of opening and vice-versa.

$$A \bullet B = (A^c \circ B)^c$$

$$A \circ B = (A^c \bullet B)^c$$

Hit – or –Miss Transform

- The hit-or-miss (HoM) transform is a combination of morphological operations that uses two structuring elements (B_1 and B_2) designed in such a way that the output image will consist of all locations that match the pixels in B_1 (a *hit*) and that have none of the pixels in B_2 (a *miss*).
- Mathematically, the HoM transform of image A by the structuring element set B ($B=(B_1.B_2)$), denoted by $A \otimes B$, is defined as

$$A \otimes B = (A \ominus B_1) \cup (A^c \ominus B_2)$$

- Alternatively, the HoM transform can be expressed as

$$A \otimes B = (A \ominus B) - (A \uparrow B_2)$$

It is the basic tool for shape detection.